

4.2 Forces: Density & Pressure

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.2 Forces: Density & Pressure
Difficulty	Hard

Time allowed: 50

Score: /34

Percentage: /100

Question 1a

The Cartesian Diver is a popular science project which can be used to demonstrate physical concepts.

A 'diver' is made by bending a loop of drinking straw into a U-shape and then weighting it with modelling clay.

A large plastic bottle is filled with water, leaving a small volume of air at the top.

The diver is carefully added, so that some water enters it but the straw remains floating upright due to trapped air.

Finally the lid is tightly screwed back onto the bottle, making a sealed container.

The completed Cartesian Diver and the construction of the diver are shown in Fig. 1.1.

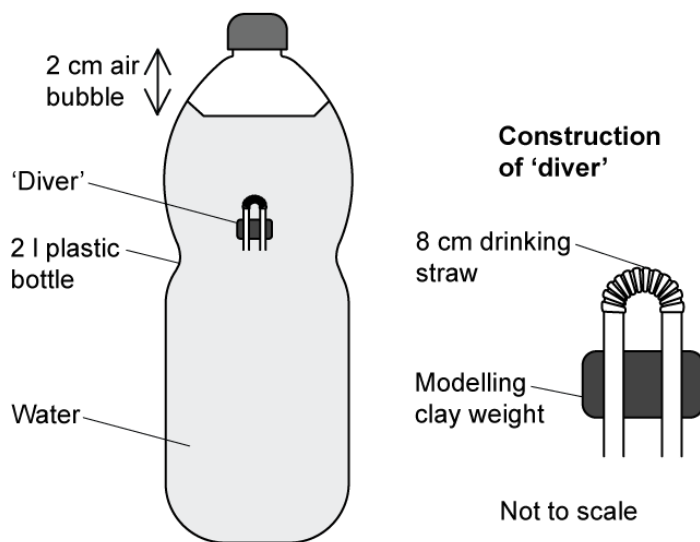


Fig. 1.1.

When the sides of the bottle are squeezed and released in response the 'diver' moves up and down rapidly as illustrated by Fig. 1.2.

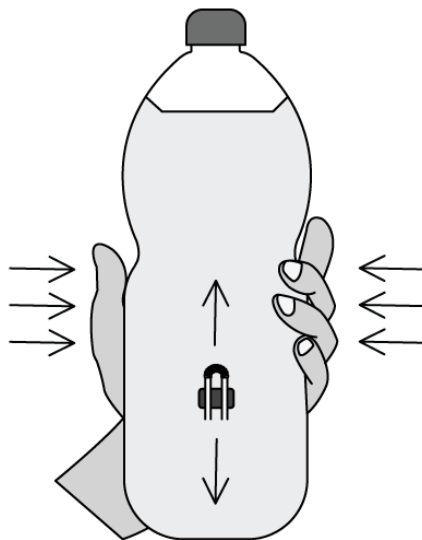


Fig. 1.2.

Discuss the physics of the Cartesian Diver when the sides of the bottle are squeezed.

(i)

Explain the observed motion of the 'diver'.

[5]

(ii)

Predict which direction the diver moves in when the sides of the bottle are squeezed.

[1]

[5 marks]

Question 1b

When the sides of the bottle are released the diver reverses direction.

Explain why the diver returns to its original position in the bottle.

[4 marks]

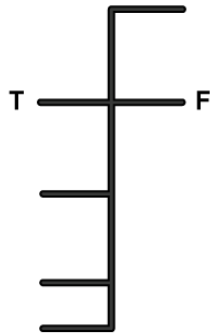
Question 2a

To prevent the overloading and sinking of ships markings, known as load lines, are required on ships hulls since the Middle Ages.

As the ship is loaded it sinks lower in the water. Once the load line meets the waterline, the ship is deemed to be fully loaded.

In 1876 Samuel Plimsoll refined the load line to reflect the needs of increasingly international trading routes. Today the 'Plimsoll line' has markings to show safe loading levels in fresh, saltwater, and warm and cold temperatures.

The Plimsoll line markings on ships today are shown in Fig. 1.1. The expected labels, their meanings and indications of temperature are written below.



F - Fresh water

T - Tropical seawater (warm)

S - Summer temperate seawater (cool)

W - Winter temperate seawater (cold)

TF - Tropical fresh water (very warm)

WNA - Winter North Atlantic (extreme cold)

Fig. 1.1.

Explain why the ships sit lower in the water when they are fully loaded.

[4 marks]

Question 2b

With reference to upthrust

(i)

Explain why different load lines for types of water and times of year are useful.

[5]

(ii)

Hence complete the diagram in Fig. 1.1. by adding the missing labels S, TF, W and WNA.

[2]

[7 marks]

Question 2c

In a simple classroom demonstration of the principle behind the Plimsoll Line, a teacher places an egg in a beaker of distilled water at close to 0°C .

The egg sinks to the bottom of the beaker as shown in Fig. 1.2.

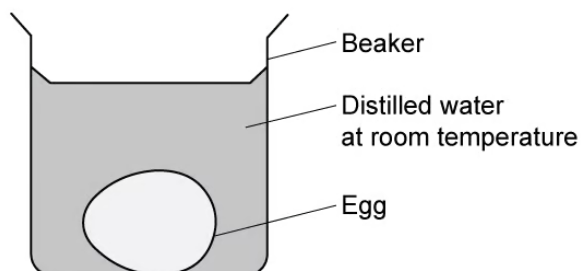


Fig. 1.2.

Suggest and comment on the likely next steps in this demonstration.

[2 marks]

Question 3a

Hot air balloons consist of an envelope with an open mouth through which air can be heated using a burner. The air inside becomes much hotter than its surroundings. Lifted by this method, the balloon can carry a large amount of weight across very long distances.

A very large capacity balloon, as shown in Fig. 1.1. can carry 25–30 people.

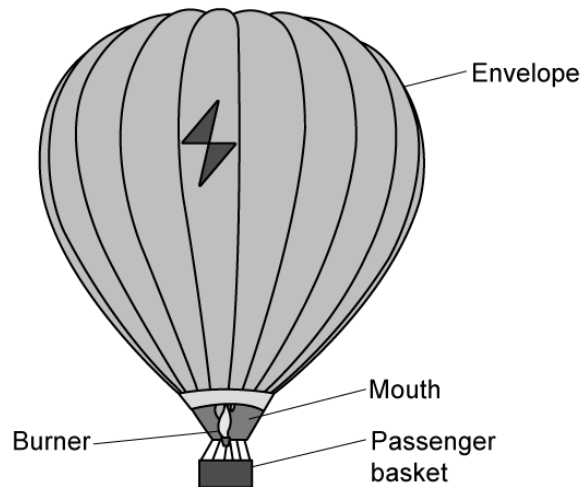


Fig. 1.1.

The envelope has a diameter at its widest point of 32 m when it is filled with air. The total mass of the balloon, air, equipment and passengers is 20 900 kg. When the air temperature is 15 °C the density of the air outside the balloon is 1.225 kg m^{-3} .

When the air inside the envelope is hot enough, the balloon is released.

(i)

Calculate the resultant upwards force on the balloon.

[3]

(ii)

Calculate the initial upwards acceleration of the balloon.

[1]

[4 marks]

Question 3b

The balloon is taken to the Middle East, where the temperature is 45°C .

Discuss how this would affect the ability of the balloon to fly and any changes the pilot could make to fly in these temperatures.

[4 marks]

Question 3c

The air density at 45°C is 4 % less than the air density at 15°C .

For the hotter conditions

(i)
Calculate the mass that must be removed in order to achieve the same initial acceleration as in part (a)(ii)

[3]

(ii)
Suggest how this might be achieved

[1]

[4 marks]



Model Answers: Hard

1a

Discuss the physics of the Cartesian Diver when the sides of the bottle are squeezed:

(i) Explain the observed motion of the 'diver':

Any **five** from

- The trapped air in the bottle compresses; [1 mark]
- Because gas is compressible (but water is not); [1 mark]
- The air pressure in the bottle increases; [1 mark]
- This increases the pressure everywhere; [1 mark]
- The bottle is a sealed unit / closed system; [1 mark]
- So, (the Increased pressure) compresses the air bubble in the diver; [1 mark]
- The density of the diver increases so it sinks; [1 mark]
- The volume of water displaced by the diver decreases; [1 mark]
- This reduces the upthrust on the diver; [1 mark]

(ii) Predict which direction the diver moves in when the sides of the bottle are squeezed:

- Downwards / sinks; [1 mark]

[Total: 6 marks]

1b

Explain why the diver returns to its original position in the bottle:

- The bottle is a closed system; [1 mark]

Any **three** from

- When the air bubble expands; [1 mark]
- The air pressure decreases; [1 mark]
- So, the compressed air bubble in the diver expands; [1 mark]

- The air is trapped; [1 mark]
- So, the diver density decreases and it moves upwards; [1 mark]

[Total: 4 marks]

2a

Explain why the ships sit lower in the water when they are fully loaded:

- (In equilibrium the) upthrust equals the weight of the ship (and its load); [1 mark]
- Upthrust is (defined as / calculated from the) weight of displaced water; [1 mark]
- When weight increases, required upthrust increases; [1 mark]
- More water has to be displaced; [1 mark]

[Total: 4 marks]

2b

i) Explain why different load lines for types of water and times of year are useful:

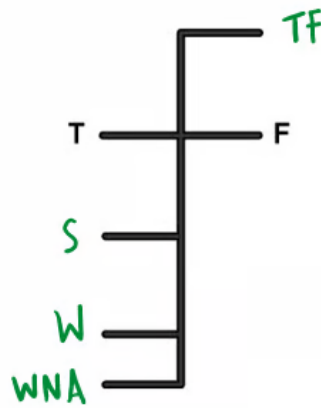
- Ships will sink to different depths with the same load (depending on the water); [1 mark]
- Because weight of displaced water must remain the same / is always the same as the weight of the ship; [1 mark]

Any **three** from (reverse argument allowed for all points, e.g.: freshwater is less dense than saltwater)

- Saltwater is more dense than freshwater; [1 mark]
- Ships are more buoyant in saltwater / saltwater provides more upthrust; [1 mark]
- Cold water is more dense than warm water; [1 mark]
- Ships are more buoyant in cold water / cold water provides more upthrust; [1 mark]

ii) Hence complete the diagram in Fig. 1.1. by adding the missing labels S, TF, W and WNA:

The correctly completed diagram looks like:



- TF and WNA correctly placed; [1 mark]
- S and W correctly placed; [1 mark]

[Total: 7 marks]

2c

Suggest and comment on the likely next steps in this demonstration:

Steps must be in this order:

- Add salt / sugar to make water more dense **AND** the egg will move upwards; [1 mark]

THEN

- (After the egg has risen) heat the water to make it less dense **AND** the egg will move downwards / sink; [1 mark]

[Total: 2 marks]

This question is harder than it looks!

The egg starts sitting on the bottom of the beaker. For the classroom demonstration to work the students must see something happen. In other words, the egg must first move upwards. Only then it can be made to move back down.

Starting with a step which would make the egg sink wouldn't work because observers wouldn't see anything happening.

Since the water is already ice cold only one method can be used to increase density. And you can't take the salt back out once it is dissolved, so the only way to reduce

density is to heat the water.

3a

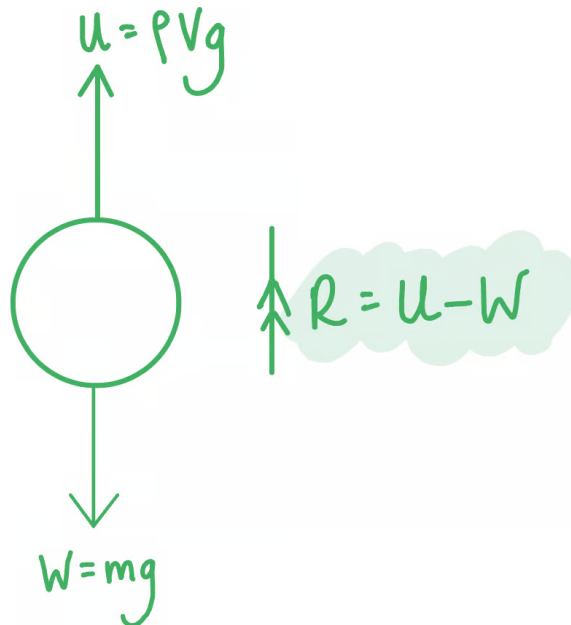
When the air inside the envelope is hot enough, the balloon is released:

(i) Calculate the resultant upwards force on the balloon:

Determine the volume of the fully inflated balloon:

- $V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \left(\frac{d}{2}\right)^3 = \frac{4}{3} \pi \times 16^3 = 17\,157 \text{ m}^3$ [1 mark]

Use a sketch to identify the forces:



Calculate upthrust:

- $U = \rho V g = 1.225 \times 17\,157 \times 9.81 = 206 \text{ kN}$ [1 mark]

Calculate weight and subtract from upthrust to find the resultant force:

- $W = m g = 20\,900 \times 9.81 = 205\,029 \text{ N}$ [1 mark]

- $R = U - W = 206\,000 - 205\,029 = 971 \text{ N}$ [1 mark]

(ii) Calculate the initial upwards acceleration of the balloon:

Apply Newton's Second Law:

- $F = ma$
- $a = \frac{R}{m} = \frac{971}{20\,900} = 0.05 \text{ m s}^{-2}$ [1 mark]

[Total: 4 marks]

3b

Discuss how this would affect the ability of the balloon to fly:

- Will be more difficult to take off; [1 mark]
- Reduce the weight the balloon is carrying; [1 mark]

Any **two** from:

- The air is less dense; [1 mark]
- Upthrust equals weight of displaced air / fluid; [1 mark]
- Less upthrust (but same load); [1 mark]

[Total: 4 marks]

3c

For the hotter conditions

(i) Calculate the mass that must be removed in order to achieve the same initial acceleration as in part (a)(ii):

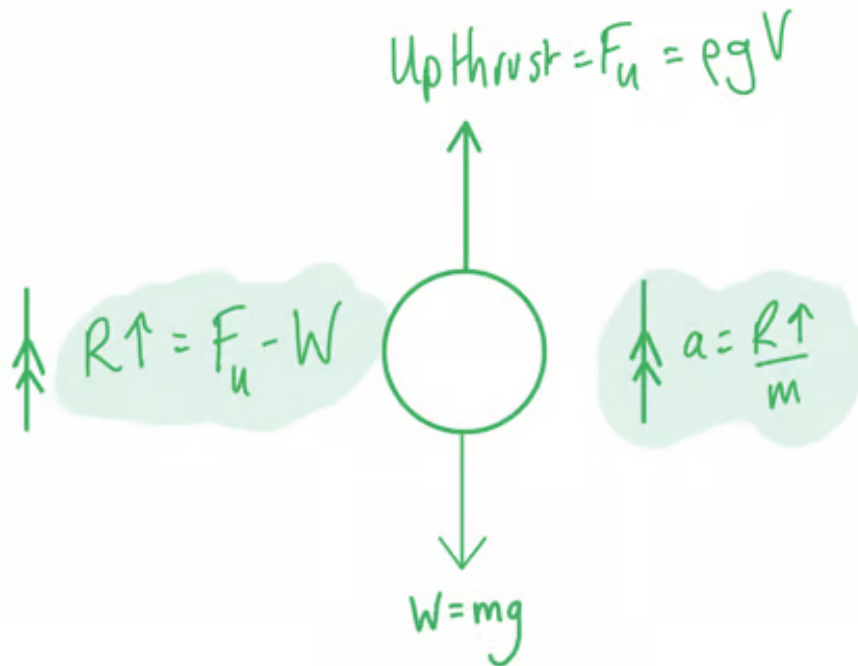
List the known quantities:

- Volume of the envelope, $V = 17\,157 \text{ m}^3$ (From part (a))
- Air density at $15^\circ\text{C} = 1.225 \text{ kg m}^{-3}$ (From part (a))
- Air density at 45°C , $\rho = 0.96 \times 1.225 \text{ kg m}^{-3} = 1.176 \text{ kg m}^{-3}$ [1 mark]

Calculate upthrust at the point of take-off:

- Upthrust, $F_u = \rho g V = 1.176 \times 9.81 \times 17\,157 = 198\,000 \text{ N}$ **or** 198 kN

Relate upthrust, weight and resultant force to calculate the resultant force using Newton's Second Law:



- $F_u - W = R \uparrow$ and $R \uparrow = ma$
- Therefore, $F_u - mg = ma$ [1 mark]
- Therefore, $F_u = ma + mg = m(a + g)$
- Maximum mass for take-off at the hotter temperature,

$$m = \frac{F_u}{a + g} = \frac{198\,000}{(0.05 + 9.81)} = 20\,081 \text{ kg [1 mark]}$$

Compare this to the original mass to find the mass that must be removed:

- $(20\,900 - 20\,081) = 820 \text{ kg (2 s.f.) [1 mark]}$

(ii) Suggest how this might be achieved:

- Reduce the number of passengers / 12–13 less passengers / smaller basket / carry less fuel; [1 mark]

[Total: 4 marks]